25th

NUMPY

In [39]:
```
import numpy as np
```

In [40]:
```
np.__version__
```

Out[40]:
```
'2.3.5'
```

In [41]:
```
np.__version__
```

Out[41]:
```
'2.3.5'
```

In [42]:
```
np.
```

```
  Cell In[42], line 1
    np.
       ^
SyntaxError: invalid syntax
```

In [43]:
```
my_list = [0,1,2,3,4,5]
my_list
```

Out[43]: [0, 1, 2, 3, 4, 5]

In [44]:
```
type(my_list)
```

Out[44]: list

In [45]:
```
arr = np.array(my_list)
arr
```

Out[45]: array([0, 1, 2, 3, 4, 5])

In [46]:
```
type(arr)
```

Out[46]: numpy.ndarray

In [47]:
```
arr
```

Out[47]: array([0, 1, 2, 3, 4, 5])

In [48]:
```
np.arange(10)
```

Out[48]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])

In [49]:
```
np.arange(10,20)
```

Out[49]: array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])

In [50]:
```
np.arange(10,20,3)
```

Out[50]: array([10, 13, 16, 19])

In [51]:
```
np.zeros(5)
```

Out[51]: array([0., 0., 0., 0., 0.])

In [52]:
```
np.zeros(5, dtype=int) #1d arry
```

Out[52]: array([0, 0, 0, 0, 0])

In [53]:
```
np.zeros((2,2)) # 2d array
```

Out[53]:
```
array([[0., 0.],
       [0., 0.]])
```

In [54]:
```
np.zeros((2,2), dtype=int)
```

Out[54]:
```
array([[0, 0],
       [0, 0]])
```

In [55]:
```
np.zeros((5,10), dtype=int)
```

Out[55]:
```
array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

In [56]:
```
np.ones(10)
```

Out[56]: array([1., 1., 1., 1., 1., 1., 1., 1., 1., 1.])

In [57]:
```
np.ones((3,3), dtype = int)
```

Out[57]:
```
array([[1, 1, 1],
       [1, 1, 1],
       [1, 1, 1]])
```

In [58]:
```
np.twos(3) #error (no 3 available)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[58], line 1
----> 1 np.twos(3) #error (no 3 available)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'twos'
```

In [59]:
```
rand(4)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[59], line 1
----> 1 rand(4)

NameError: name 'rand' is not defined
```

In [60]:
```
random.rand(4)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[60], line 1
----> 1 random.rand(4)

NameError: name 'random' is not defined
```

In [61]:
```
np.random.rand(4) #execute multiple time it will different
```

Out[61]:
```
array([0.11569273, 0.71578278, 0.44125163, 0.53863654])
```

In [62]:
```
np.random.rand(4,4)
```

Out[62]:
```
array([[0.83736661, 0.98389401, 0.87150587, 0.8423558 ],
       [0.98214168, 0.14860014, 0.18327731, 0.02605126],
       [0.70157994, 0.48308339, 0.9954078 , 0.51789545],
       [0.68027847, 0.62294444, 0.25856938, 0.67629711]])
```

In [63]:
```
np.random.rand(2,2) #everytime we generating float
```

Out[63]:
```
array([[0.34168371, 0.70710288],
       [0.70514978, 0.59869507]])
```

In [64]:
```
np.random.randint(3)
```

Out[64]:
```
0
```

In [65]:
```
np.random.randint(10, 15)
```

Out[65]: 12

In [66]:
```
np.random.randint(1,5,3)
```

Out[66]: array([4, 3, 2], dtype=int32)

In [68]:
```
np.random.randint(1,5,4)
```

Out[68]: array([4, 1, 2, 1], dtype=int32)

In [69]:
```
np.random.randint(1,5)
```

Out[69]: 1

In [71]:
```
np.random.randint(5,2,3) # 1st no must be smaller than 2nd
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[71], line 1
----> 1 np.random.randint(5,2,3)

File numpy/random/mtrand.pyx:794, in numpy.random.mtrand.RandomState.randint()

File numpy/random/_bounded_integers.pyx:1425, in numpy.random._bounded_integers._rand_int32()

ValueError: low >= high
```

In [73]:
```
np.random.randint(-5,2,3) # first no must be smaller than 2nd
```

Out[73]: array([-1, -3, -4], dtype=int32)

In [80]:
```
np.random.randint(10,40,(10,10))
```

Out[80]:
```
array([[14, 32, 13, 27, 33, 37, 33, 25, 19, 11],
       [20, 20, 37, 37, 25, 30, 28, 18, 10, 32],
       [26, 26, 32, 35, 12, 30, 32, 16, 19, 12],
       [26, 11, 20, 14, 23, 23, 14, 38, 22, 28],
       [18, 35, 26, 18, 22, 27, 29, 26, 36, 29],
       [34, 29, 26, 15, 23, 39, 17, 30, 30, 27],
       [15, 18, 28, 12, 38, 34, 26, 13, 12, 33],
       [32, 28, 35, 34, 17, 38, 36, 30, 38, 34],
       [36, 11, 32, 37, 13, 17, 33, 36, 37, 11],
       [10, 28, 37, 21, 14, 26, 22, 15, 39, 13]], dtype=int32)
```

In [82]:
```
ri =  np.random.randint(10,40,(10,10))
ri
```

Out[82]:
```
array([[33, 25, 15, 21, 33, 31, 11, 19, 17, 10],
       [20, 28, 23, 39, 15, 18, 36, 34, 35, 27],
       [24, 38, 31, 13, 19, 23, 39, 36, 29, 27],
       [13, 27, 34, 12, 17, 22, 24, 25, 21, 30],
       [25, 35, 11, 35, 22, 14, 14, 29, 17, 28],
       [29, 30, 23, 19, 19, 13, 17, 23, 17, 33],
       [28, 12, 27, 12, 31, 12, 26, 11, 38, 39],
       [20, 23, 39, 13, 27, 21, 30, 15, 17, 28],
       [20, 22, 31, 19, 23, 17, 38, 22, 29, 12],
       [15, 11, 12, 24, 34, 35, 15, 14, 37, 24]], dtype=int32)
```

In [83]:
```
ri =  np.random.randint(10,40,(3,5))
ri
```

Out[83]:
```
array([[15, 29, 30, 10, 26],
       [15, 29, 23, 23, 31],
       [18, 16, 16, 17, 36]], dtype=int32)
```

In [84]:
```
len(ri) # it will print bydefault 10 rows
```

Out[84]:
```
3
```

In [86]:
```
ri.shape # it will give rows and colomns
```

Out[86]:
```
(3, 5)
```

In [87]:
```
id(ri) # address of the memory location
```

Out[87]:
```
2336592053808
```

In [88]:
```
ri
```

Out[88]:
```
array([[15, 29, 30, 10, 26],
       [15, 29, 23, 23, 31],
       [18, 16, 16, 17, 36]], dtype=int32)
```

Slicing in Numpy

In [89]:
```
ri[:] # it will print all
```

Out[89]:
```
array([[15, 29, 30, 10, 26],
       [15, 29, 23, 23, 31],
       [18, 16, 16, 17, 36]], dtype=int32)
```

In [90]:
```
ri[1]
```

Out[90]:
```
array([15, 29, 23, 23, 31], dtype=int32)
```

In [91]:
```
ri
```

Out[91]:
```
array([[15, 29, 30, 10, 26],
       [15, 29, 23, 23, 31],
       [18, 16, 16, 17, 36]], dtype=int32)
```

In [92]:
```
ri[0:3]
```

Out[92]:
```
array([[15, 29, 30, 10, 26],
       [15, 29, 23, 23, 31],
       [18, 16, 16, 17, 36]], dtype=int32)
```

In [93]:
```
ri_1 = np.random.randint(10,40,(10,10))
ri_1
```

Out[93]:
```
array([[38, 23, 30, 38, 24, 24, 17, 15, 22, 37],
       [23, 31, 24, 10, 32, 14, 16, 10, 35, 25],
       [28, 17, 14, 23, 20, 23, 33, 17, 22, 32],
       [19, 13, 35, 13, 30, 13, 12, 11, 26, 23],
       [24, 24, 24, 39, 10, 32, 29, 32, 24, 38],
       [23, 14, 20, 13, 10, 29, 31, 37, 23, 19],
       [28, 36, 17, 33, 12, 38, 20, 15, 23, 19],
       [35, 35, 14, 30, 36, 24, 10, 10, 39, 25],
       [23, 27, 28, 18, 19, 38, 13, 13, 38, 36],
       [31, 37, 37, 14, 16, 39, 12, 15, 14, 13]], dtype=int32)
```

In [94]:
```
ri_1.shape
```

Out[94]:
```
(10, 10)
```

In [95]:
```
ri_1 = np.random.randint(10,40,(10,10))  #dtype=int32 will not show
print(ri_1)
```

```
[[19 28 17 37 17 14 17 34 27 29]
 [15 35 38 21 35 27 15 39 38 37]
 [37 12 21 15 17 32 24 32 34 21]
 [37 24 31 27 16 33 18 22 29 35]
 [26 15 28 37 23 11 14 38 28 27]
 [26 13 18 22 33 13 24 23 17 15]
 [37 24 27 16 16 10 14 39 22 36]
 [10 10 30 24 33 28 27 12 30 34]
 [39 29 16 25 15 32 28 18 15 12]
 [32 13 17 16 19 20 24 34 24 39]]
```

In [96]:
```
ri_1
```

Out[96]:
```
array([[19, 28, 17, 37, 17, 14, 17, 34, 27, 29],
       [15, 35, 38, 21, 35, 27, 15, 39, 38, 37],
       [37, 12, 21, 15, 17, 32, 24, 32, 34, 21],
       [37, 24, 31, 27, 16, 33, 18, 22, 29, 35],
       [26, 15, 28, 37, 23, 11, 14, 38, 28, 27],
       [26, 13, 18, 22, 33, 13, 24, 23, 17, 15],
       [37, 24, 27, 16, 16, 10, 14, 39, 22, 36],
       [10, 10, 30, 24, 33, 28, 27, 12, 30, 34],
       [39, 29, 16, 25, 15, 32, 28, 18, 15, 12],
       [32, 13, 17, 16, 19, 20, 24, 34, 24, 39]], dtype=int32)
```

In [97]:
```
ri_1[3:8]
```

Out[97]:
```
array([[37, 24, 31, 27, 16, 33, 18, 22, 29, 35],
       [26, 15, 28, 37, 23, 11, 14, 38, 28, 27],
       [26, 13, 18, 22, 33, 13, 24, 23, 17, 15],
       [37, 24, 27, 16, 16, 10, 14, 39, 22, 36],
       [10, 10, 30, 24, 33, 28, 27, 12, 30, 34]], dtype=int32)
```

In [99]:
```
ri_1
```

Out[99]:
```
array([[19, 28, 17, 37, 17, 14, 17, 34, 27, 29],
       [15, 35, 38, 21, 35, 27, 15, 39, 38, 37],
       [37, 12, 21, 15, 17, 32, 24, 32, 34, 21],
       [37, 24, 31, 27, 16, 33, 18, 22, 29, 35],
       [26, 15, 28, 37, 23, 11, 14, 38, 28, 27],
       [26, 13, 18, 22, 33, 13, 24, 23, 17, 15],
       [37, 24, 27, 16, 16, 10, 14, 39, 22, 36],
       [10, 10, 30, 24, 33, 28, 27, 12, 30, 34],
       [39, 29, 16, 25, 15, 32, 28, 18, 15, 12],
       [32, 13, 17, 16, 19, 20, 24, 34, 24, 39]], dtype=int32)
```

In [100…

```
ri_1[3,8] # always count from  0
```

Out[100…

```
np.int32(29)
```

In [101…

```
ri_1
```

Out[101…

```
array([[19, 28, 17, 37, 17, 14, 17, 34, 27, 29],
       [15, 35, 38, 21, 35, 27, 15, 39, 38, 37],
       [37, 12, 21, 15, 17, 32, 24, 32, 34, 21],
       [37, 24, 31, 27, 16, 33, 18, 22, 29, 35],
       [26, 15, 28, 37, 23, 11, 14, 38, 28, 27],
       [26, 13, 18, 22, 33, 13, 24, 23, 17, 15],
       [37, 24, 27, 16, 16, 10, 14, 39, 22, 36],
       [10, 10, 30, 24, 33, 28, 27, 12, 30, 34],
       [39, 29, 16, 25, 15, 32, 28, 18, 15, 12],
       [32, 13, 17, 16, 19, 20, 24, 34, 24, 39]], dtype=int32)
```

In [102…

```
ri_1[-5]
```

Out[102…

```
array([26, 13, 18, 22, 33, 13, 24, 23, 17, 15], dtype=int32)
```

In [103…

```
ri_1[-5,-3] ddddddddd
```

Out[103…

```
np.int32(23)
```

In [105…

```
ri_1
```

Out[105…

```
array([[19, 28, 17, 37, 17, 14, 17, 34, 27, 29],
       [15, 35, 38, 21, 35, 27, 15, 39, 38, 37],
       [37, 12, 21, 15, 17, 32, 24, 32, 34, 21],
       [37, 24, 31, 27, 16, 33, 18, 22, 29, 35],
       [26, 15, 28, 37, 23, 11, 14, 38, 28, 27],
       [26, 13, 18, 22, 33, 13, 24, 23, 17, 15],
       [37, 24, 27, 16, 16, 10, 14, 39, 22, 36],
       [10, 10, 30, 24, 33, 28, 27, 12, 30, 34],
       [39, 29, 16, 25, 15, 32, 28, 18, 15, 12],
       [32, 13, 17, 16, 19, 20, 24, 34, 24, 39]], dtype=int32)
```

In [106…

```
ri_1[::3]
```

Out[106…

```
array([[19, 28, 17, 37, 17, 14, 17, 34, 27, 29],
       [37, 24, 31, 27, 16, 33, 18, 22, 29, 35],
       [37, 24, 27, 16, 16, 10, 14, 39, 22, 36],
       [32, 13, 17, 16, 19, 20, 24, 34, 24, 39]], dtype=int32)
```

In [107…

```
ri_1
```

Out[107…

```
array([[19, 28, 17, 37, 17, 14, 17, 34, 27, 29],
       [15, 35, 38, 21, 35, 27, 15, 39, 38, 37],
       [37, 12, 21, 15, 17, 32, 24, 32, 34, 21],
       [37, 24, 31, 27, 16, 33, 18, 22, 29, 35],
       [26, 15, 28, 37, 23, 11, 14, 38, 28, 27],
       [26, 13, 18, 22, 33, 13, 24, 23, 17, 15],
       [37, 24, 27, 16, 16, 10, 14, 39, 22, 36],
       [10, 10, 30, 24, 33, 28, 27, 12, 30, 34],
       [39, 29, 16, 25, 15, 32, 28, 18, 15, 12],
       [32, 13, 17, 16, 19, 20, 24, 34, 24, 39]], dtype=int32)
```

In [108…

```
ri_1[::-1]
```

Out[108…

```
array([[32, 13, 17, 16, 19, 20, 24, 34, 24, 39],
       [39, 29, 16, 25, 15, 32, 28, 18, 15, 12],
       [10, 10, 30, 24, 33, 28, 27, 12, 30, 34],
       [37, 24, 27, 16, 16, 10, 14, 39, 22, 36],
       [26, 13, 18, 22, 33, 13, 24, 23, 17, 15],
       [26, 15, 28, 37, 23, 11, 14, 38, 28, 27],
       [37, 24, 31, 27, 16, 33, 18, 22, 29, 35],
       [37, 12, 21, 15, 17, 32, 24, 32, 34, 21],
       [15, 35, 38, 21, 35, 27, 15, 39, 38, 37],
       [19, 28, 17, 37, 17, 14, 17, 34, 27, 29]], dtype=int32)
```

In [109…

```
ri_1[::-5] # backward print
```

Out[109…

```
array([[32, 13, 17, 16, 19, 20, 24, 34, 24, 39],
       [26, 15, 28, 37, 23, 11, 14, 38, 28, 27]], dtype=int32)
```

In [110…

```
ri_1
```

Out[110…

```
array([[19, 28, 17, 37, 17, 14, 17, 34, 27, 29],
       [15, 35, 38, 21, 35, 27, 15, 39, 38, 37],
       [37, 12, 21, 15, 17, 32, 24, 32, 34, 21],
       [37, 24, 31, 27, 16, 33, 18, 22, 29, 35],
       [26, 15, 28, 37, 23, 11, 14, 38, 28, 27],
       [26, 13, 18, 22, 33, 13, 24, 23, 17, 15],
       [37, 24, 27, 16, 16, 10, 14, 39, 22, 36],
       [10, 10, 30, 24, 33, 28, 27, 12, 30, 34],
       [39, 29, 16, 25, 15, 32, 28, 18, 15, 12],
       [32, 13, 17, 16, 19, 20, 24, 34, 24, 39]], dtype=int32)
```

In [111…
```
ri_1[2]
```

Out[111…
```
array([37, 12, 21, 15, 17, 32, 24, 32, 34, 21], dtype=int32)
```

In [113…
```
print(ri_1[2]) # specific row and specific coloumn
print(ri_1[:,2])
```

```
[37 12 21 15 17 32 24 32 34 21]
[17 38 21 31 28 18 27 30 16 17]
```

In [114…
```
ri_1
```

Out[114…
```
array([[19, 28, 17, 37, 17, 14, 17, 34, 27, 29],
       [15, 35, 38, 21, 35, 27, 15, 39, 38, 37],
       [37, 12, 21, 15, 17, 32, 24, 32, 34, 21],
       [37, 24, 31, 27, 16, 33, 18, 22, 29, 35],
       [26, 15, 28, 37, 23, 11, 14, 38, 28, 27],
       [26, 13, 18, 22, 33, 13, 24, 23, 17, 15],
       [37, 24, 27, 16, 16, 10, 14, 39, 22, 36],
       [10, 10, 30, 24, 33, 28, 27, 12, 30, 34],
       [39, 29, 16, 25, 15, 32, 28, 18, 15, 12],
       [32, 13, 17, 16, 19, 20, 24, 34, 24, 39]], dtype=int32)
```

In [116…
```
ri_1[:, 8] # always count from 0
```

Out[116…
```
array([27, 38, 34, 29, 28, 17, 22, 30, 15, 24], dtype=int32)
```

In [117…
```
ri_1[:, -8]  #count from back
```

Out[117…
```
array([17, 38, 21, 31, 28, 18, 27, 30, 16, 17], dtype=int32)
```

In [118…

```
ri_1
```

Out[118…

```
array([[19, 28, 17, 37, 17, 14, 17, 34, 27, 29],
       [15, 35, 38, 21, 35, 27, 15, 39, 38, 37],
       [37, 12, 21, 15, 17, 32, 24, 32, 34, 21],
       [37, 24, 31, 27, 16, 33, 18, 22, 29, 35],
       [26, 15, 28, 37, 23, 11, 14, 38, 28, 27],
       [26, 13, 18, 22, 33, 13, 24, 23, 17, 15],
       [37, 24, 27, 16, 16, 10, 14, 39, 22, 36],
       [10, 10, 30, 24, 33, 28, 27, 12, 30, 34],
       [39, 29, 16, 25, 15, 32, 28, 18, 15, 12],
       [32, 13, 17, 16, 19, 20, 24, 34, 24, 39]], dtype=int32)
```

In [119…

```
ri_1[0:3, 0:3]
```

Out[119…

```
array([[19, 28, 17],
       [15, 35, 38],
       [37, 12, 21]], dtype=int32)
```

In [120…

```
np.arange(1,13)
```

Out[120…

```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12])
```

In [121…

```
np.arange(1,13).reshape(3,4)
```

Out[121…

```
array([[ 1,  2,  3,  4],
       [ 5,  6,  7,  8],
       [ 9, 10, 11, 12]])
```

In [122…

```
np.arange(1,13).reshape(5,4) # 5 * 4 = 10 but we have 12 here thats y we get error
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[122], line 1
----> 1 np.arange(1,13).reshape(5,4)

ValueError: cannot reshape array of size 12 into shape (5,4)
```

In [123…

```
np.arange(1,13).reshape(2,6)
```

Out[123…

```
array([[ 1,  2,  3,  4,  5,  6],
       [ 7,  8,  9, 10, 11, 12]])
```

In [124…

```
np.arange(1,13).reshape(1,12)
```

Out[124…

```
array([[ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12]])
```

numpy masking or filter

In [125…

```
mat = np.arange(0,100).reshape(10,10)
mat
```

Out[125…

```
array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
       [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
       [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
       [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
       [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

In [126…

```
id(mat)
```

Out[126…

```
2336592179152
```

In [127…

```
mat.shape
```

Out[127…

```
(10, 10)
```

In [128…

```
mat
```

Out[128…

```
array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
       [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
       [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
       [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
       [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

In [130…

```
mat < 50
```

Out[130…

```
array([[ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [False, False, False, False, False, False, False, False, False,
        False],
       [False, False, False, False, False, False, False, False, False,
        False],
       [False, False, False, False, False, False, False, False, False,
        False],
       [False, False, False, False, False, False, False, False, False,
        False],
       [False, False, False, False, False, False, False, False, False,
        False]])
```

In [131…

```
mat[mat < 50 ]
```

Out[131…

```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
       17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
       34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

In [132…

```
mat[mat == 50 ]
```

Out[132…

```
array([50])
```

In [133…

```
mat[mat != 50 ]
```

Out[133…

```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
       17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
       34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 51,
       52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68,
       69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85,
       86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99])
```

In [134…

```
mat[mat>50]
```

Out[134…

```
array([51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67,
       68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84,
       85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99])
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: